

HENRI AÏDASSO

Research Scientist | Artificial Intelligence and Software Engineering
Ph.D. @ École de technologie supérieure (ÉTS)
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RESEARCH INTERESTS

Machine Learning, Reliability of Multi-Component Systems, AI System Reliability and Evaluation, Low-Resource NLP, AI for Software Engineering, Efficient Machine Learning Methods.

EDUCATION

Ph.D. in Computer Engineering École de technologie supérieure (ÉTS) Thesis: Intelligent Detection and Diagnosis of Anomalies in Software Delivery Pipelines Grade: Excellent, GPA: 4.3/4.3	2022 — Mar 2026 Montreal, Canada
M.Sc. in Big Data: Decision Intelligence and Machine Learning University of Rennes 1 Grade: Valedictorian, with highest honors “ <i>Mention Très Bien</i> ”	2020 — 2022 Rennes, France
B.Sc. in Computer Science University of Rennes 1 Grade: Valedictorian, with highest honors “ <i>Mention Très Bien</i> ”	2019 — 2020 Rennes, France
B.Sc. in Computer Science and Management University of Abomey-Calavi (UAC) Grade: Valedictorian, <i>Trophy of Excellence</i>	2014 — 2017 Cotonou, Benin

EXPERIENCES

Research and Development Scientist Computer Research Institute of Montreal (CRIM) • Part-time contract	Aug 2025 — Apr 2026 Montreal, Canada
- Optimized preprocessing and inference pipelines for statistical downscaling of spatiotemporal climate variables using U-Net models (ouranosinc/resoterre), achieving 30% latency reduction. - Contributed to open-source geospatial data standardization (stac-utils/pystac), improving data quality and accessibility for climate science ML training pipelines, in collaboration with UoT and Ouranos.	
Applied Research Scientist TELUS • Mitacs Accelerate Fellowship	May 2023 — Apr 2026 Montreal, Canada
- Led applied research on AI for software reliability, building NLP and statistical models for intermittent and silent failures detection, root cause diagnosis, and failure prioritization. Reduced infrastructure costs by 25%, addressing failure categories estimated at over \$1.5M. - Developed data-efficient predictive models for anomaly detection, achieving 88% F1-score with only 12 labeled examples per class, validated across 45,000+ Monte Carlo experimental configurations. Designed a novel log search algorithm inspired by binary search, reducing developer investigation time by 74.4%. - Collaborated with engineers to translate the research findings into a deployed AI-powered framework.	

Data Scientist

Energiency • Full-time contract

Sep 2020 — Aug 2022

Rennes, France

- Developed and maintained the cloud-based ML backend enabling distributed training and real-time inference for dozens of production models, and created reusable data science libraries adopted across teams.
- Developed anomaly detection models on production line sensor data (*DS Smith*) to predict paper breakage events, reducing unplanned downtime and associated costs.
- Conducted data science studies on industrial energy consumption, applying time series forecasting (XGBoost, LSTM) and what-if analysis to sensor data, improving energy efficiency by 15% across Industry 4.0 manufacturing clients.

Software Engineer

Worldline Global • Internship

May 2020 — Aug 2020

Rennes, France

Developed automated financial monitoring tools for *Tap2Use*, the leading digital payment solution for transportation in France, producing daily analytics over large data volumes across distributed databases.

Software Engineer

Arriauz Services Santé • Contract

Nov 2018 — Oct 2020

Cotonou, Benin

Developed and maintained *Dis-moi Doc*, a web and mobile health services platform featuring instant doctor chat and drug search across 5 countries. 15,000+ downloads on the Google Play Store in 2020.

Software Developer

Contracts and Internships

Sep 2016 — Oct 2018

Cotonou, Benin

Android and full-stack web developer consultant on various projects in healthcare, finance, education, and business management across seven governmental and software organizations.

SELECTED PUBLICATIONS

Predicting Intermittent Job Failure Categories for Diagnosis Using Fine-Tuned Language Models

Henri Aïdasso, Francis Bordeleau, Ali Tizghadam

Proceedings of the ACM International Conference on the Foundations of Software Engineering (FSE) Industry Track. July 2026. Just accepted, pages to appear.

Towards Open-Ended Discovery for Low-Resource NLP

Bonaventure F. P. Dossou, Henri Aïdasso

Proceedings of the 2nd Workshop on Uncertainty-Aware NLP (UncertaiNLP) @ EMNLP 2025. September 2025, pp. 287-297.

Efficient Detection of Intermittent Job Failures Using Few-Shot Learning

Henri Aïdasso, Francis Bordeleau, Ali Tizghadam

Proceedings of the IEEE 41st International Conference on Software Maintenance and Evolution (ICSME). September 2025, pp. 632-643.

On the Diagnosis of Flaky Job Failures: Understanding and Prioritizing Failure Categories

Henri Aïdasso, Francis Bordeleau, Ali Tizghadam

Proceedings of the IEEE/ACM 47th International Conference on Software Engineering: Software Engineering in Practice (ICSE SEIP). May 2025, pp. 192-202.

Towards Build Optimization Using Digital Twins

Henri Aïdasso, Francis Bordeleau, Ali Tizghadam

Proceedings of the ACM 21st International Conference on Predictive Models and Data Analytics in Software Engineering (PROMISE). June 2025, pp. 95-98.

SKILLS

Research (4+ yrs): Empirical Studies, Design Science Research, Mixed-Methods Research, Experimental Design, Statistical Modeling, Qualitative Analysis, Literature Review. **Machine Learning** (6+ yrs): Classical ML, Deep Learning, NLP, Information Retrieval, LLMs, Embeddings, Fine-Tuning (Few-Shot Learning), RAG, AI Agents, Time Series Modeling. **Programming**. Python (8 yrs), Java (6 yrs), Node.js (4 yrs), R (2 yrs), Bash (2 yrs), C++ (1 yr). **Software Engineering** (8+ years): Backend Development (REST APIs, Libraries, CLIs), Distributed and Concurrent Systems, System Design, Automated Testing, Telemetry and Chaos Engineering, CI/CD. **MLOps** (6+ years): Containerization (Docker), Kubernetes Orchestration, Cloud Services (AWS), Model Serving and Monitoring, **Data Systems** (8 yrs): Distributed Training (GPU clusters), Data Modeling, Big Data Processing (Apache Spark). **Mathematics**: Probability, Statistics, Linear Algebra, Mathematical Analysis.

Frameworks: PyTorch, Transformers, scikit-learn, SetFit, LangChain, LangGraph, FastAPI, smolagents.
Database Systems: PostgreSQL, Vector Databases (Qdrant, FAISS), MongoDB.

HONORS AND AWARDS

Nominee for the Doctoral Excellence Award	2026
Thesis graded Excellent (Honor List) and nominated for the ÉTS Doctoral Excellence Award	
Arbour Foundation Doctoral Scholarship	2025
Highly selective \$30,000 CAD scholarship awarded to outstanding doctoral researchers in Quebec universities for academic excellence, leadership and social impact.	
Mitacs Accelerate Fellowship	2023 – 2026
Industry-academia research collaboration with TELUS (\$72,000 over 3 years).	
Top of Class / Valedictorian	2022, 2020
Highest french academic distinction (Mention <i>Très Bien</i>) across Master's and Bachelor's degrees.	
Trophy of Excellence	2017
Awarded for achieving Top of Class all three years (2015, 2016, 2017) of the Bachelor's degree.	
National Mathematics Champion	2014
Won national championship. Represented Benin at the International Mathematical Olympiad (IMO).	

TEACHING AND SUPERVISION EXPERIENCES

Lecturer Graduate Research Supervisor	Sum 2025, Win 2026
École de technologie supérieure (ÉTS) · Contract	Montreal, Canada
· Lecturer for LOG680 — Reliable Automation and Software Analytics.	
· Supervised M.Sc. and M.Eng. projects on process modeling and agentic AI for reliability engineering.	

SERVICE AND COMMUNITY INVOLVEMENT

Academic Service: Program Committee: ACM/IEEE MSR 2026, IEEE CASCON 2025, UncertainNLP @ EMNLP 2025 | Reviewer: ACM/IEEE MODELS 2024, ACVSS 2024–2025.

University (ÉTS): Chair, Organizing Committee of the Annual Congress of Student Researchers (CAéC 2026) | Coordinator, Graduate Student Committees of the AÉÉTS (since W25) | Mentor, Contact Program for international students (F25) | Student Member, Doctoral Excellence Award Committee (F25, W26).

LANGUAGES

French: Native Language | **English**: Professional working proficiency | **Fon**: Native Language